Passkeys.Tools

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Content

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Passkeys.Tools

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Passkeys.Tools

Encode, Decode, Intercept, Modify, and Exploit WebAuthn Operations

Passkeys.Tools is a comprehensive security testing and development toolkit for WebAuthn (passkey) implementations. It provides full emulation of both the client (browser) and authenticator layers, allowing security researchers, penetration testers, and developers to analyze relying party implementations for vulnerabilities and compliance issues.

Key Capabilities: Decode and inspect WebAuthn attestations and assertions • Craft credentials with any algorithm • Intercept and modify WebAuthn API calls • Test challenge binding, signature verification, and origin validation • Simulate cross-session attacks between multiple browser profiles • Verify compliance with the WebAuthn specification.

**Standalone Mode

Offline analysis and manual crafting of WebAuthn data

A passive toolkit for decoding, modifying, and encoding WebAuthn attestations and assertions—think jwt.io but for passkeys. All data (keys, users, history) is stored locally in your browser's localStorage.

Use when: You have captured WebAuthn data (e.g., from network logs) and need to analyze or modify it offline.

Example: Decode an attestation object to inspect the credential public key, modify the signature counter, then re-encode it for replay testing.

**Interceptor Mode

Live interception and modification of WebAuthn API calls

Builds on Standalone Mode by adding real-time interception via the browser extension. The extension hooks into the WebAuthn API to capture and modify all `navigator.credentials.create()` and `get()` calls before they reach the relying party—similar to Burp Suite but specialized for WebAuthn. Data remains in localStorage. Use passthrough to forward the request to your browser's or device's native authenticator (Touch ID, Windows Hello, etc.). The response is captured and displayed for inspection or modification before sending back to the relying party.

Use when: You need to test a relying party's server-side validation by modifying live WebAuthn responses.

Example: Intercept a registration, replace the challenge with random bytes, and verify if the server properly validates challenge binding. Or flip a bit in the signature to test signature verification.

**Cross-Browser Mode

Multi-browser testing with shared remote storage

Extends Interceptor Mode with a remote storage backend, enabling two browser profiles (e.g., victim and attacker) to share keys, users, and history in real-time. This is essential for testing cross-session attacks where data must flow between separate browser instances. Optional end-to-end encryption protects your data on the server.

Use when: You need to simulate attacks involving multiple parties, such as session binding or credential reuse across different sessions.

Example: In browser A (victim), start a passkey registration and dismiss it. In browser B (attacker), intercept a registration and swap in the victim's challenge. If the server accepts this, it indicates a session binding vulnerability.

**Create

Trigger `navigator.credentials.create()` API calls with custom PublicKeyCredentialCreationOptions. Test how the browser and authenticator handle different registration parameters, algorithms, and attestation preferences.

**Get

Trigger `navigator.credentials.get()` API calls with custom PublicKeyCredentialRequestOptions. Test authentication flows with specific credential IDs, user verification requirements, and mediation settings.

****Attestation**

Decode, inspect, modify, and re-encode attestation objects and clientDataJSON. Supports `none` and `packed` attestation formats. Manipulate flags, AAGUID, credential IDs, public keys, and extensions.

****Assertion**

Decode, modify, and encode assertion authenticatorData, clientDataJSON, and signatures. Sign assertions using stored private keys or verify existing signatures against stored public keys.

****Keys**

Manage cryptographic keys with full import/export support. Generate key pairs for all WebAuthn algorithms (ES256, ES384, ES512, RS256, PS256, EdDSA, etc.). Associate credential IDs with keys for realistic credential emulation.

****Users**

Store and manage user account data including RP IDs, usernames, display names, and user handles. Users are automatically captured during registration interceptions for easy reuse in authentication tests.

****History**

Complete audit log of all intercepted operations. Search, filter, and export entries. Copy values between sessions to perform cross-operation attacks like challenge swapping or credential reuse.

****Converters**

Encoding utilities for Base64, Base64URL, and Hex conversions. Convert keys between JWK, COSE, PEM, and DER formats for compatibility with different tools and libraries.

****Interceptor**

Central hub for live WebAuthn interception (requires browser extension). View request details, select credentials and keys, apply one-click security tests (challenge manipulation, signature tampering, flag modifications), and craft custom responses. Use passthrough to forward requests to your native authenticator and capture real responses for inspection or modification.

Follow these step-by-step instructions to set up your preferred usage mode:

****Standalone Mode Setup**

No setup required! Simply open Passkeys.Tools in your browser and start using the Attestation, Assertion, Keys, and Converters tabs. All data is automatically stored in your browser's localStorage and persists across sessions.

****Interceptor Mode Setup**

To intercept live WebAuthn operations, you need to install and configure the browser extension:

- **Install the extension:** Install the extension from the Chrome Web Store.

- **Configure the extension:** Click the extension icon in your browser toolbar to open the popup. Configure these settings:
- **Frontend URL:** The URL of your Passkeys.Tools instance (default: `https://passkeys.tools`). Change this if you're self-hosting.
- **Operation Mode:** Select *Default*. Each website gets its own credential scoped to that website (normal behavior).
- **Popup Display Mode:** Choose *Detached Window* (opens in a separate 1200x800 window) or *Inline Popup* (opens as an overlay).
- **Storage:** Data is stored in your browser's localStorage by default. No additional configuration needed.
- **Start intercepting:** Visit any website that uses WebAuthn passkeys. When a registration or authentication is triggered, Passkeys.Tools will open automatically, allowing you to inspect and modify the request before responding.

****Cross-Browser Mode Setup**

To share data between multiple browser profiles or browsers, you need to configure the extension for profile mode and enable remote storage:

- **Install the extension** from the Chrome Web Store in all Chrome profiles you want to use.
- **Configure Operation Mode:** In the extension popup, select *Profile 1* in one browser (e.g., attacker) and *Profile 2* in another (e.g., victim). All websites will share the same credential scoped to the selected profile, enabling cross-browser credential swapping tests.
- **Enable Remote Storage:** Navigate to the Settings tab and select *Remote Storage* mode.
- **Set the Server URL:** Enter the URL of your storage backend:
- *Hosted backend:* Use our provided backend at `https://db.passkeys.tools` (no setup required).
- *Self-hosted:* Run your own backend server and enter its URL.
- **Configure a Secret:** Enter a unique secret string. This secret:
- Acts as your "account identifier" on the backend—all browsers using the same secret share the same data.
- Must be identical across all browsers/profiles you want to sync.
- Should be kept private; anyone with your secret can access your stored data.
- **Important:** If you lose your secret while using E2E encryption, your encrypted data cannot be recovered. Store your secret securely.
- **Enable End-to-End Encryption (recommended):** When enabled, all data is encrypted in your browser before being sent to the server. The server only stores encrypted blobs and cannot read your keys, users, or history. The encryption key is derived from your secret and never leaves your browser.
- **Apply settings in all browsers:** Use the exact same Server URL, Secret, and E2E Encryption setting in every browser profile that should share data.

- **Verify sync:** Create a test key in one browser, then check the Keys tab in another browser—it should appear automatically.

| Shortcut | Action | |

Create Passkey

Get Passkey

clientDataJSON

attestationObject

authenticatorData

publicKey

clientDataJSON

authenticatorData

signature

Keys

| Name | Credential ID | Public Key | Private Key | |

Users

| RP ID | Name | Display Name | User ID | Mode | |

General Converters

Key Converters

Interceptor

****Overview**

| Mode | - | | | Type | - | | | Origin | - | | | Cross Origin | - | | | Top Origin | - | | | Mediation | - | |

****Actions**

No actions available

****Attacks**

No attacks available

Settings

Choose your preferred color theme.

Dark Mode

Automatic Match system preference

Light Always use light theme

Dark Always use dark theme

Save Theme

Choose how to store your keys, users, and history.

Mode

Local Storage Store data in browser's local storage

Remote Storage Store data on a remote server

Save Storage

History

| Timestamp | Info | Origin | Credential / Key / User | Actions | |
